Plants and Society

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Plants and Society (A01002)

Short Title: Plants and Society
Faculty Science and Computing
Department Science
Cluster Horticulture
Author CDALY
Credits 5 **Level:** Intermediate

Description of Module / Aims

This module examines the importance of plants in modern society and tackles some of the increasingly complex issues that surround the biotechnology of plants. It is necessary for science students to understand these issues and to practice defending their informed opinions, the protocols used and also the plant science research that takes place during the development of new biotechnologies. Society increasingly looks to science graduates to bridge the gap in their understanding of novel biotechnology and this module will help science students to develop skills in evaluating news sources and debating, and ultimately, will produce graduates who will inform society and foster trust in the scientific process. This elective is offered to any science student in WIT who seeks to develop their knowledge of plant biotechnology, the importance of plants in society, and who wants to learn how to generate convincing arguments.

Programmes

HORT-0034	BSc in Horticulture (WD_SHORT_D)	3 / 6 / E
HORT-0034	BSc in Horticulture (WD_SHORT_DX)	3 / 6 / E

Indicative Content

- Climate Change: The role of plants in carbon sequestration and carbon release. Early earth environments. Climate change through the eons Plant fossils. Global average temperatures. The greenhouse effect.
- Evolution: Fundamentals of evolution and speciation. Cladistics and the tree of life. Evidence for Evolution. Evolution controversies. Antibiotic use in plant propagation and evolution in the context of antibiotic resistance
- Food security: Fundamentals of GM technology. Terminator technology. Land use change. Rising populations. Intergovernmental policies regarding food security
- Science communication: Distinguishing between reliable sources of information. Responsible use of social media, blogs and message boards. Using evidence based reasoning to participate in constructive and informative arguments
- Nature deficit disorder: Specifically, the reasoning behind why society may not appreciate plant life
- Popular Science: Evaluating the quality of popular science and methods of incorporating popular science in life-long learning.

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Explore reliable scientific information from a variety of print and online resources.
2. Appraise popular science publications from a variety of print and online resources.
3. Investigate the on-going environmental impacts of fertiliser, antibiotic and GM plant use.
4. Analyse the process of inquiry and the assimilation of knowledge.
5. Establish how plants have contributed to and can be used to mitigate global climate change.
6. Research and discuss an appropriate topic.
7. Produce a researched and informative presentation on a given topic.
8. Express opinions logically and persuasively in group discussions.

Learning and Teaching Methods

- Lectures
- Workshops
- Moodle-based Discussion Forums

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Group Project	40%	1,2,3,5,6
Participation	40%	4,8
Presentation	20%	7

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in research ability/ skill and understanding to carry out practical tasks. Unwilling to participate in group discussions.

40%-49%: Demonstrates a limited knowledge of the subject matter. Show some research ability/skill when presenting evidence-based arguments.

50%-59%: Demonstrates satisfactory general knowledge of the main issues discussed. Logically and competently carries out research tasks and presents evidence adequately.

60%-69%: Demonstrates sound knowledge of the subject matter. Presentations and reports completed to high standard. Participates adequately in class discussions.

70%-100%: Demonstrates excellent research ability and awareness of the complexities of modern-day plant science and demonstrates an aptitude for delivering convincing arguments using evidences based reasoning. Regularly directs discussions and encourages peer discussion.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Workshop	24	
Independent Learning	87	

Requested Resources

- Lecture Room: Loose Seated